

```
info = 1
octave:3>
```

So, the number of flowers the sage came with is 5 and each offering is of 15 flowers.

Note that in all these solutions, the trick is to select the initial solution close to the actual solution, through some approximation. At times that might be tricky. In case we just have polynomial equations (and that also in one variable), they can be solved in an easier way, using the polynomial features of Octave in contrast to the earlier method. Here, we also get all of the multiple solutions for the polynomial.

Playing around with polynomials

Let's consider the polynomial $2x^3 + 3x^2 + 2x + 1 = 0$. Then, Octave's representation and computation of its solutions or roots would be as follows:

```
$ octave -qf
octave:1> P = [2; 3; 2; 1];
octave:2> roots(P)
ans =
```

```
-1.00000 + 0.00000i
-0.25000 + 0.66144i
-0.25000 - 0.66144i
```

```
octave:3>
```

Being a cubic equation, it has three roots, as expected. The first one is the real number -1, and the other two are complex conjugates $(-1 + \sqrt{-7})/4$ and $(-1 - \sqrt{-7})/4$. And you may verify the solutions using the function `polyval` as follows:

```
octave:1> P = [2; 3; 2; 1];
octave:2> sols = [-1; (-1 + sqrt(-7)) / 4; (-1 + sqrt(-7)) / 4]
sols =
```

```
-1.00000 + 0.00000i
-0.25000 + 0.66144i
-0.25000 + 0.66144i
```

```
octave:3> polyval(P, sols)
ans =
```

```
0
0
0
```

```
octave:4>
```

This shows that the value of the polynomial P evaluated at each of the three solutions is 0. Hence, confirming that they indeed are the solutions.

Being all set with the basics of polynomials in Octave, let's solve some puzzles.

Geometry

Last month, we found the point of intersection of a straight line and a circle. We only calculated one point, though typically there should be two. It would be only one in case of the straight line being a tangent to or just touching the circle. It would be zero if the straight line does not intersect the circle. Let's now try these different cases, with the one variable polynomial power.

Let us have the following circle C with radius 5 and centred at origin $(0, 0)$, defined in the Cartesian coordinate system, i.e., the x - y system: $x^2 + y^2 = 25$

Let us consider the following three lines intersecting the above circle, one by one:

$$L1: 4x + 3y = 24$$

$$L2: x + y = 5\sqrt{2}$$

$$L3: 6x + y = 36$$

To be able to solve the points of intersection of each of these three lines with the circle C using `roots`, the first step is to get polynomials in one variable. For that, we can substitute the value of y in the equation of the circle, in terms of x , from each of the line equations, as follows:

For $L1$

$$x^2 + y^2 \text{ fl } 25 \quad \square \quad 9x^2 + 9y^2 \text{ fl } 9 \cdot 25 \quad \square \quad 9x^2 + (24 - 4x)^2 \text{ fl } 225 \quad \square \quad 25x^2 - 192x + 351 = 0$$

For $L2$

$$x^2 + y^2 \text{ fl } 25 \quad \square \quad x^2 + (5\sqrt{2} - x)^2 \text{ fl } 25 \quad \square \quad 2x^2 - 10\sqrt{2}x + 25 = 0$$

For $L3$

$$x^2 + y^2 \text{ fl } 25 \quad \square \quad x^2 + (36 - 6x)^2 \text{ fl } 25 \quad \square \quad 37x^2 - 432x + 1271 = 0$$

We get the roots of each to get the x co-ordinate of the point of intersection:

```
$ octave -qf
octave:1> C1 = [25; -192; 351];
octave:2> C2 = [2; -10*sqrt(2); 25];
octave:3> C3 = [37; -432; 1271];
octave:4> roots(C1)
ans =
```

```
4.6800
3.0000
```

```
octave:5> roots(C2)
ans =
```